The Macro Anchor as a Short-Rate Central Tendency: Bond Risk Premia, Out-of-Sample Discipline, and a Specification Audit*

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July 2026 — first draft

Abstract

This paper asks whether the intuition that nominal yields gravitate toward a *macro-consistent anchor* — the neutral real rate plus trend inflation — contains tradeable information about bond risk premia, and answers it by escalating the discipline of the test rather than the flexibility of the signal. Two natural formulations fail honestly: long-yield levels barely mean-revert, and the raw 10-year "anchor gap," despite strong in-sample significance on a 50-year sample (Newey–West $t = 2.86$) and cleanly monotone return quintiles, does not survive out of sample ($R^2_{OOS} = -0.49$; Cochrane–Piazzesi fares worse). The idea survives only in its theoretically disciplined form: the anchor as the *central tendency of a Hull–White short-rate process*, with reversion estimated at the front end where it exists and the term premium obtained as an *output*. The resulting model term premium is not the slope in disguise (correlation 0.35), tracks the Kim–Wright benchmark (0.70) with three transparent parameters, is significant in sample at every tenor with t-statistics monotone in maturity (2.06 to 3.87), and is the only predictor in a five-way horse race with a *positive realized* out-of-sample R^2 (basket +0.012, Clark–West 10%; exact 10Y target +0.085, Clark–West 5%). A pre-specified replication across nine developed markets yields 9/9 positive slopes and positive R^2_{OOS} in 6/9. The paper's distinctive section is a *specification and data audit* that decomposes the entire $0.012 \rightarrow 0.215$ spread across test variants into named, quantified ingredients — tenor concentration (+0.07), a duration-approximated target (+0.05), the funding-rate series (+0.07), and sample length (+0.06) — and shows the reversion parameter is identified by pre-ZLB history. Results are robust to a publication-lag inflation input. The framing throughout is *theory as regularization*: the signal with the best in-sample fit has the worst out-of-sample record, and the model-disciplined form is the only survivor.

Keywords: bond risk premia; term premium; affine short-rate models; out-of-sample R^2; Clark–West; neutral rate; specification audit.

*Working paper. All results are produced by fully reproducible, deterministic notebooks (expanding-window estimation throughout; FRED/OECD data with documented fallbacks). All errors are mine. Comments welcome.

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1 Introduction

A recurring intuition in macro-rates investing is that nominal yields should gravitate not toward their own historical mean but toward a *macro-consistent anchor*: the neutral real short rate r^* plus expected long-run inflation π_e (plus, for long tenors, a term premium). When the observed yield sits far above that anchor, bonds look cheap against fundamentals; far below, rich. The intuition is old, respectable, and dangerous — because it invites exactly the kind of in-sample overfitting that the return-predictability literature has learned to distrust (Goyal and Welch, 2008; Bauer and Hamilton, 2018).

This paper takes the intuition seriously by doing the opposite of tuning it. We fix the anchor's construction once, test the idea in three progressively more disciplined formulations, report the failures with the same prominence as the success, and subject the surviving result to a replication across nine markets and to a full specification and data audit. Four findings emerge.

First, two honest failures. As a *level-timing* signal the idea is dead on arrival: at monthly frequency the 10-year level and slope are statistically indistinguishable from random walks (estimated reversion speeds $\kappa \approx 0.02$ and ≈ 0). As a *raw risk-premium* predictor — regressing next-12-month bond excess returns on the gap between the 10-year yield and the anchor, in the spirit of Cieslak and Povala (2015) — the signal is strongly significant in sample on a five-decade sample (Newey–West $t = 2.86$, $R^2 = 0.13$, and mean subsequent excess returns rising monotonically from +0.13% to +6.55% across gap quintiles) yet posts an out-of-sample R^2 of -0.49 against the expanding-mean benchmark of Campbell and Thompson (2008). For context, the Cochrane and Piazzesi (2005) forward-rate factor does worse (-1.24) on the same protocol. This is the Goyal and Welch (2008) pattern in miniature, and we report it as a result, not a nuisance.

Second, the idea survives in exactly one form — the theoretically disciplined one. We embed the anchor where the data say reversion actually lives: as the time-varying central tendency $\theta_t = r_t^* + \pi_{e,t}$ of a Hull–White short-rate process (Hull and White, 1990), estimated under the physical measure by regressing Δr on the gap to the anchor — the project's original "policy-gap" intuition promoted to a parameter estimator. The pure-expectations curve (expectations plus convexity, zero price of risk) then delivers a *model-implied term premium per tenor as an output*, $TP(\tau, t) = y_{obs}(\tau, t) - y_{PE}(\tau, t)$. Three honesty checks discipline the object: the HW term premium correlates only 0.35 with the slope (constant-θ Vasicek and CIR analogues, lacking the anchor, correlate 0.77 and 0.89 — they *are* the slope in disguise); it tracks the Kim–Wright five-factor benchmark at 0.70 with three transparent parameters (the anchor-free models track it at 0.03 and 0.18); and its in-sample predictive t-statistics are monotone in tenor, from 2.06 at two years to 3.87 at ten.

Third, the surviving signal passes the out-of-sample gate that everything else fails, and replicates across markets. In a five-way horse race on a common evaluation window (1997–2026, 336 months) the HW term premium is the only predictor with a positive *realized* out-of-sample R^2 (+0.012 on a 2–10Y basket; +0.085 on the exact 10-year target), with Clark–West significance at the 10% and 5% level respectively; slope and Fama–Bliss have comparable Clark–West statistics but negative realized R^2_{OOS} — population content destroyed by estimation noise — and lose their 5% stars once evaluated on the common window. Running the *same,* *pre-specified* pipeline on nine developed markets yields positive slopes in 9/9 countries, $R^2_{OOS} > 0$ in 6/9, and Clark–West above the 10% threshold in 7/9, with the failures (Australia, Norway, a too-short Swiss sample) reported alongside.

Fourth — and the paper's distinctive contribution — an audit, not just a result. The panel's US row is far stronger ($R^2_{OOS} = +0.215$) than the strict US benchmark (+0.012). Rather than headline the larger number, we decompose the entire gap with a reconciliation ladder and a month-by-month data audit: tenor concentration accounts for +0.07 (the effect lives at the long end; the basket dilutes it), the duration-approximated target for +0.05 (it shares the carry with the predictor and omits roll-down), the funding-rate series for +0.07 (T-bill versus interbank — the raw ten-year yields are *numerically identical* across sources, so the residual is entirely the short leg), and the longer sample for +0.06. The signal itself is identical across specifications (the cross-variants coincide to machine precision). A sample-cut diagnostic adds a structural finding: re-estimating on post-1997 data alone drives the reversion speed to its floor ($\kappa \to 10^{-4}$) and the result collapses — the anchor parameter is *identified by pre-ZLB history*; the zero-rate decade alone contains no information about it. A pre-specified reading rule, written before the audit numbers, then fixes the framing: the panel is long-sample cross-country replication; the strict benchmark is the US common-window result.

The paper relates to the bond risk-premium literature descending from Fama and Bliss (1987) and Cochrane and Piazzesi (2005), to the trend-inflation "cycle" factor of Cieslak and Povala (2015) — our raw gap is essentially their object, and our failure/success pattern refines where its content resides — and to the out-of-sample skepticism of Goyal and Welch (2008), Bauer and Hamilton (2018) and Duffee (2002). Methodologically we lean on Campbell and Thompson (2008) and Clark and West (2007) for evaluation, on Newey and West (1987) for inference with overlapping targets, on Vasicek (1977), Cox et al. (1985) and Hull and White (1990) for the model, on Holston et al. (2017) and Kim and Wright (2005) for the anchor's external references, and on Pesaran and Smith (1995) for the panel mean-group estimate. Our contribution is not a new predictor class; it is a demonstration, unusually complete in its accounting, of *where* a popular macro intuition holds, where it does not, and how much of any headline number is specification.

2 Data and the macro anchor

United States. Monthly, June 1976 to May 2026 (599 observations). Constant-maturity Treasury yields GS1–GS10 (FRED); the state variable is the 3-month T-bill (TB3MS); CPI is CPIAUCSL; the external term-premium reference is the Kim–Wright 10-year series (THREEFYTP10). The 1976 start is imposed by GS2 availability. All series are month-end.

The anchor. We deliberately freeze a simple, replicable construction:

$$\theta_t \;=\; r^*_t + \pi_{e,t}, \qquad \pi_{e,t} = \mathrm{EWMA}_{36m}(\Delta_{12} \log \mathrm{CPI}), \qquad r^*_t = \mathrm{EWMA}_{120m}(r_t - \pi_{e,t}), \tag{1}$$

i.e. trend inflation is a three-year-half-life average of trailing year-on-year CPI (the construction underlying the Cieslak and Povala, 2015 cycle), and the neutral real rate is a slow ten-year-half-life average of the realized real short rate (the Holston–Laubach–Williams estimate is used instead whenever reachable; the notebook prints which source was used). No term-premium block enters the *short-rate* anchor: for long tenors the term premium is an output of the model, not an input. Section 8 shows the results are essentially unchanged when CPI enters with a one-month publication lag.

G10 panel. Nine markets (USA, DEU, GBR, CAN, CHE, SWE, NOR, AUS, NZL; Japan is dropped for insufficient overlapping data — a convenient absence we flag, since Japan is the archetypal ZLB case where the model is least at home). Sources are OECD MEI series on FRED: 10-year government yields (`IRLTLT01*`), 3-month interbank rates (`IR3TIB01*`), CPI (`*CPIALLMINMEI`, quarterly fallback for AUS/NZL). Spans range from 1960 (DEU, GBR, CAN) to 1999 (CHE). Absent full foreign curves, the panel target is the standard duration approximation $rx^{(10)}_{t\to t+12} \approx y^{(10)}_t - D(y)\,\Delta y^{(10)} - r^{3m}_t$; the audit of Section 7 quantifies exactly how much this construction flatters results.

3 Two honest failures

Level timing. Estimated at monthly frequency, the reversion speeds of the first two principal components of the US curve are $\kappa \approx 0.022$ (level) and ≈ 0 (slope): half-lives of decades. Whatever the anchor knows, it is not that the ten-year yield will fall soon. The allocation experiments that motivated this project (not reported here; available in the replication repository) confirm that carry and roll-down, not level reversion, do the work in curve strategies.

The raw risk-premium gap. Reframed as a return predictor, the natural but ultimately failed version is the deviation of the 10-year yield from a legacy long-yield anchor $r^* + \pi_e + TP$ regressed on next-12-month bond excess returns. This is not the short-rate anchor used below: here the term-premium block is an external long-yield benchmark component, whereas in the disciplined specification the term premium is derived as a model output. On 1976–2026 the in-sample evidence is genuinely strong: Newey–West $t = 2.86$ (lag 12), $R^2 = 0.13$, and mean subsequent 12-month excess returns by gap quintile of $\{+0.13, +1.9, +3.4, +4.7, +6.55\}$ percent — clean monotonicity from "rich" to "cheap." Out of sample, against the expanding-mean benchmark with expanding-window re-estimation: $R^2_{OOS} = -0.49$. The Cochrane–Piazzesi forward factor, evaluated identically, posts -1.24. On the modern subsample (2004–2026) the gap's in-sample t-statistic is only 1.18 with $R^2_{OOS} = -0.14$. We record both failures at full prominence: the in-sample strength is concentrated in the 1980s disinflation and does not transfer to a real-time forecaster. The natural response is not to tune the signal but to ask *where* the economics actually operate.

4 The anchor as a short-rate central tendency

Reversion is real at the front end: policy drags the short rate toward neutral. We therefore place the anchor where the reversion lives, as the central tendency of the instantaneous rate, and derive the whole curve.

Table 1: Full-sample physical-measure estimates, US 3-month rate, 1976–2026 (annualized).

Model	κ	half-life (months)	θ	σ
OU / Vasicek	0.094	88.8	3.87%	1.49%
CIR (NCχ^2 ML)	0.053	157	3.57%	0.061
HW (anchor)	0.175	47.5	$r_t^* + \pi_{e,t}$ (moving)	1.48%

Notes: the CIR Feller condition is violated ($2\kappa\theta = 0.0019 < \sigma^2 = 0.0034$), consistent with a decade at the zero bound. AR-based $\hat{\kappa}$ is biased upward near a unit root. The HW gap-regression t-statistic is 1.37: the halved half-life relative to the constant-θ OU is a point estimate consistent with the thesis, not a statistically decisive one — we say so.

Table 2: Is the model term premium the slope in disguise? Does it track the benchmark?

	corr(TP_{10}, slope)	corr(TP_{10}, Kim–Wright)
HW (anchor)	**0.35**	**0.70**
OU (constant θ)	0.77	0.03
CIR	0.89	0.18

Notes: the anchor is the only ingredient distinguishing HW from OU. Without it, the "term premium" is a repackaged slope and tracks nothing; with it, a three-parameter transparent model tracks the Fed's five-factor benchmark at 0.70.

4.1 Models and estimation under the physical measure

Three diffusions for the 3-month rate r_t: Ornstein–Uhlenbeck/Vasicek $dr = \kappa(\theta - r)dt + \sigma dW$ (Gaussian transition, closed form); CIR $dr = \kappa(\theta - r)dt + \sigma\sqrt{r}\,dW$ (non-central χ^2 transition); and Hull–White with the *macro anchor as moving central tendency*, $dr = \kappa(\theta_t - r)dt + \sigma dW$ with $\theta_t = r_t^* + \pi_{e,t}$. Estimation is always under the physical measure P from the time series — never calibrated to today's curve, so the comparison with the observed curve is informative rather than circular. OU is fit by its exact AR(1) discretization; CIR by exact-moment weighted least squares, refined by non-central-χ^2 maximum likelihood; HW by regressing Δr_t on the gap $(\theta_t - r_t)$ — the slope *is* $\kappa\Delta$. All downstream signals use *expanding-window* parameters (120-month burn-in); full-sample estimates (Table 1) are descriptive only.

4.2 The term premium as an output, and three honesty checks

For each model the pure-expectations yield (expectations plus convexity, zero price of risk) is closed-form affine, $y_{PE}(\tau) = (-\ln A(\tau) + B(\tau)\,r_t)/\tau$ with $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$; the HW variant evaluates the Vasicek formula at $\theta = \theta_t$ (a random-walk forecast of a slow anchor, stated rather than hidden). The model term premium is the wedge $TP_m(\tau, t) = y_{obs}(\tau, t) - y^m_{PE}(\tau, t)$. Table 2 reports the checks that decide whether this object is new information.

5 Predictive evidence

Targets are per-tenor 12-month excess returns in the Fama–Bliss construction, $rx^{(n)}_{t\to t+12} = n\,y_t(n) - (n-1)\,y_{t+12}(n-1) - y_t(1)$ (off-grid maturities interpolated), and a 2–10Y equal-weight basket.

Table 3: In-sample predictive regressions: $rx^{(n)} \sim TP_{HW}(n)$, Newey–West lag 12.

Tenor	2y	3y	5y	7y	10y
β per +1SD (%/12m)	+0.25	+0.48	+1.24	+2.08	+3.48
NW t	2.06	2.10	2.80	3.35	**3.87**
R^2	0.035	0.037	0.076	0.117	0.172

Notes: basket $t = 3.04$, $R^2 = 0.087$. The monotone tenor structure is documented here, *before* any out-of-sample claim; it licenses reporting the exact 10Y result in Table 4 as structure rather than selection.

Table 4: Horse race on the common evaluation window (1997:05–2026:05, $n = 336$ for all).

Predictor	R^2_{OOS}	Clark–West t	significance
TP_{HW} (anchor model), basket	+0.012	1.31	10%
TP_{HW}, exact 10Y target	+0.085	2.25	**5%**
TP_{OU} (constant θ)	−0.080	1.19	—
Slope ($y_{10} - y_1$)	−0.010	1.60	10%
Fama–Bliss forward spread	−0.015	1.62	10%
Raw anchor gap (legacy)	−0.468	0.51	—

Notes: on their longer native windows, slope and Fama–Bliss carry 5% Clark–West stars ($t = 1.66, 1.74$) with *negative* realized R^2_{OOS} — population-level content eaten by estimation noise; both stars fail to survive the common window. The HW term premium is the only predictor whose realized forecasts beat the benchmark. Adding the slope to TP_{HW} *worsens* R^2_{OOS} (−0.066): the information is in the model object.

Inference uses Newey–West with lag 12 (overlapping targets). Out-of-sample R^2 follows Campbell and Thompson (2008): expanding forecasting regressions against the expanding mean. Nested significance uses Clark and West (2007) with NW(12) errors, one-sided.

The claims hierarchy we defend is therefore: (i) the pre-specified strict benchmark — basket $R^2_{OOS} = +0.012$, Clark–West 10%; (ii) the documented tenor structure — exact 10Y $R^2_{OOS} = +0.085$, Clark–West 5%; and, next, (iii) cross-country replication with magnitudes explicitly discounted by the audit of Section 7.

6 Cross-country replication

The identical pipeline — same anchor recipe and half-lives, same HW gap regression, same expanding windows, same tests, nothing re-tuned — is run on nine developed markets. Because nothing is chosen on the new data, this stage is better interpreted as a frozen-pipeline replication exercise than as another draw from a specification search. Cross-country dependence and heterogeneous sample spans still warrant caution, so the sign-replication count is treated as the primary panel statistic.

7 The specification and data audit

The panel's US row ($R^2_{OOS} = +0.215$) is an order of magnitude stronger than the strict US benchmark (+0.012). A result this sensitive to specification is not reportable until the sensitivity

Table 5: G10 replication of the pre-specified pipeline (duration-approximated 10Y target).

	β/SD	NW t	R^2_{IS}	R^2_{OOS}	CW t	n_{OOS}
USA	0.047	5.64	0.29	+0.216	4.74	467
CAN	0.036	5.18	0.20	+0.222	4.69	520
SWE	0.032	4.27	0.21	+0.201	3.20	197
DEU	0.025	3.42	0.11	+0.016	1.73	520
NOR	0.024	3.18	0.14	−0.080	1.65	221
AUS	0.017	2.00	0.04	−0.112	−0.49	404
NZL	0.027	1.90	0.07	+0.110	3.06	350
CHE	0.019	1.78	0.11	−0.429	0.38	47
GBR	0.014	1.50	0.03	+0.023	1.79	520

Signs: **9/9** positive. NW $t > 2$: 5/9 (> 1.65: 8/9). $R^2_{OOS} > 0$: **6/9**. CW > 10% crit.: 7/9.
Mean-group $\beta = +2.66\%$/SD, $t = 7.85$ (Pesaran and Smith, 1995) — overstated by cross-country dependence; the sign-replication count is the primary statistic, as pre-specified. Failures reported: AUS, NOR; CHE has only 47 evaluations; JPN dropped for data overlap (the ZLB case — its absence is flagged, not hidden).

Table 6: Reconciliation ladder, US data, common window (1997:05–, $n = 336$).

Pair	R^2_{OOS}	CW t	NW t
A1: basket signal → basket exact	+0.012	1.31	3.04
A2: 10Y signal → 10Y exact	+0.085	2.25	3.87
B1: panel signal → 10Y dur.-approx.	+0.140	2.70	4.31
B2: 10Y signal → 10Y dur.-approx.	+0.140	2.70	4.31
B3: panel signal → 10Y exact	+0.085	2.25	3.87

Notes: B1=B2 and B3=A2 exactly — the signal is invariant; the target explains everything on-sample.

itself is measured. We do so in three steps, all on identical evaluation months, with reading rules written *before* the numbers.

7.1 A reconciliation ladder: target versus signal

On the same US data and common window we evaluate five pairs: (A1) basket signal → basket exact target; (A2) 10Y signal → exact 10Y target; (B1) panel-style signal → duration-approximated 10Y target; and the cross-variants (B2) affine signal on the approximated target, (B3) panel signal on the exact target. Table 6 shows B1≡B2 and B3≡A2 *to machine precision*: the panel's signal is numerically the same object as the affine notebook's — so the entire on-sample gap is the **target**, never the signal. The decomposition is two steps: basket → exact 10Y adds +0.07 of R^2_{OOS} (tenor concentration, licensed by Table 3); exact → duration-approximated adds +0.05. The approximation flatters for two identifiable reasons: it shares the yield level (carry) mechanically with the predictor, and, lacking roll-down, aging and cross-tenor noise, it is a smoother target that a slow signal explains more easily. It is the standard construction of the international bond-premium literature, but the honest hierarchy is exact-Fama–Bliss first.

7.2 A row-by-row data audit: the funding series

What remains between the affine-data B1 (+0.135 on common rows) and the panel-USA row is data source and sample. A month-by-month comparison of every ingredient over the common 1976–2025 span (586 months) finds: the ten-year yields are *numerically identical* (the OECD long rate is the Fed's GS10; correlation 1.000, zero difference); CPI is identical to a few basis points. The entire source residual is the **3-month rate**: T-bill versus interbank, mean difference −53bp, spiking to −365bp in October 2008 (the TED blowout) and to $-250/-316$bp in 1980–82 — and it propagates into the r^* proxy (correlation drops to 0.70, also via EWMA burn-in state), the anchor (−94bp mean), the model TP (+67bp) and the target's funding leg. On *identical rows* the same formula yields $R^2_{OOS} = 0.135$ (CW 2.67) with the T-bill and 0.208 (CW 4.17) with the interbank rate: **seven points of out-of-sample R^2 hinge on the choice of the short-rate series**. We report both and pick neither; there is a defensible case for the interbank rate as the state variable (a smoother funding rate, free of T-bill flight-to-quality distortions), and an equally standard case for the T-bill as the correct funding leg for Treasury excess returns. The final ingredient, extending the sample from 1976 back to 1964, is worth roughly +0.06.

7.3 Sample cuts and parameter identification

Re-running the panel pipeline on US windows: from 1976 the result survives ($R^2_{OOS} = +0.153$, CW 3.03) — it is not a pre-1976 artifact. Re-estimating on post-1997 data *alone*, it collapses ($R^2_{OOS} = -0.061$, $n = 72$) with the reversion speed at its numerical floor ($\hat{\kappa} = 10^{-4}$). The distinction with the strict benchmark matters: the benchmark *learns* κ from 1976 onward and *evaluates* from 1997 (and survives); the cut both learns and evaluates post-1997. The economics: **the anchor-reversion parameter is identified by the pre-ZLB history** — a decade pinned at zero contains no information about reversion toward a moving neutral rate. The pre-specified rule then fixes the framing: the panel is long-sample cross-country replication; the US common-window result is the strict benchmark.

7.4 Accounting for the full gap

Adding the pieces: tenor concentration +0.07, target approximation +0.05, funding series +0.07, sample +0.06 — the entire spread from the strict +0.012 to the panel US +0.215 is attributed to named, quantified specification choices, with the signal itself invariant. To our knowledge such an accounting is rarely reported in this literature; we suggest it should be standard whenever a predictability claim spans several test designs.

8 Robustness and the distributional layer

Publication-lag inflation. Shifting CPI by one month (the real-time proxy) leaves the strict benchmark essentially unchanged: R^2_{OOS} +0.0093 versus +0.0116, Clark–West 1.32 versus 1.31, NW t 3.07. The 36-month-half-life trend makes the signal slow by construction; this verifies it. Full real-time treatment (ALFRED vintages, revisions and not just lag) is the stated upgrade, not claimed.

Calibration of the transition densities. Probability-integral-transform tests of the 12-month-ahead short-rate distribution (expanding parameters) favor the level-dependent-volatility CIR over the Gaussian OU (KS distance 0.093 versus 0.209; overlapping observations, so shapes rather than stars). An honest negative rides along: in a variance-forecast race, trailing realized variance dominates both models' closed-form variances (R^2_{OOS} 0.685 versus 0.013 for CIR and -0.087 for OU) — the "closed forms win on second moments" conjecture fails against simple persistence. The two models' 1% lower quantiles diverge sharply as rates approach zero (maximum disagreement September 2011), a clean model-based ZLB regime flag.

Economic value. Sizing duration on the model's closed-form $\mathbb{E}[rx]$ and $SD[rx]$ (constant-TP forecast for the aged yield; signals lagged one month) improves on static duration: Sharpe $0.58 \to 0.71$–0.75 for sign/linear sizing with maximum drawdown halved ($-19.7\% \to -8/-11\%$); the z-sized book is the most defensive (vol 2.3%, MaxDD -6.1%, Sharpe 0.63). Gross of transaction costs.

9 Discussion: theory as regularization, and multiple testing

Two patterns organize the results. First, *theory as regularization*: the raw anchor gap has the best in-sample fit ($R^2 = 0.128$) and the worst out-of-sample record (-0.47); the model-disciplined form has less in-sample fit and is the only out-of-sample survivor. Imposing the process structure — expectations, convexity, reversion placed where it is identified — disciplines the signal the way a prior disciplines a regression. Second, an explicit multiple-testing account: this is the *third* formulation of one idea tested on US data (the two failures are reported at equal prominence), so the marginal significance levels deserve a further mental discount (Harvey et al., 2016); the G10 stage, being a frozen pipeline on new data, does not add to the count. We regard the resulting claim — "survives out of sample in its disciplined form, with a marginal star, replicated in sign across nine markets, with every specification sensitivity quantified" — as the strongest *honest* statement the data permit, and deliberately weaker than the strongest statement the numbers would superficially allow.

10 Limitations

One-factor models cannot fit a three-factor curve; part of any model TP is misfit and is treated relative to its own history. $\hat{\kappa}$ is biased upward near a unit root, and its gap-regression t-statistic (1.37) is suggestive, not decisive. The CIR Feller condition is violated in a ZLB sample, and the Gaussian HW admits negative rates. The duration-approximated panel target flatters (+0.05, quantified); panel data are revised OECD series with heterogeneous spans; cross-country observations are not independent. Results are sensitive to the funding-rate series (± 0.07; both reported). Economic value is gross of costs. The evaluation windows, while common where it matters, remain a single historical path.

11 Conclusion

A popular macro intuition — yields gravitate toward r^* plus trend inflation — contains real information about bond risk premia, but only when embedded where its economics operate: as the central tendency of the short rate, with the term premium derived rather than assumed. Tested with pre-specified protocols, the disciplined form is the only predictor among standard benchmarks whose realized out-of-sample forecasts beat the expanding mean, replicates in sign across nine markets, survives a real-time inflation proxy, and comes with a complete specification audit that attributes every point of variation across test designs. The failures along the way — level timing, the raw gap, the variance-forecast conjecture, three countries out of sample, a decade that identifies nothing — are reported at the same volume as the successes, because in this literature the reporting standard *is* the result.

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